FINAL PROJECT REPORT: GUESS THE NUMBER VISUAL GAME

Course: Programming II

GitHub Repository: https://github.com/EmreGuzel0/GuessTheNumber.git

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1. Project Overview

"Guess The Number Visual Game" is an interactive, JavaFX-based application designed as the final assessment for the Programming II course. The application provides a gamified experience where the computer generates a random target number, and the player must identify it using logic and feedback hints.

The project strictly adheres to the mandated technologies: Java as the primary language and JavaFX as the GUI framework.

2. Requirement Compliance & Scoring

This project was developed to meet all four scoring categories defined in the project guidelines:

- **Basic Functions (50 Points):** Includes dynamic number generation, guess validation, and game-state management (Play Again, Difficulty toggles).
- **Authentication (10 Points):** A complete system for User Registration, Login, and Logout functionality.
- **File Processing (10 Points):** Permanent storage of user credentials and high scores on the local drive.
- **Presentation & Documentation (30 Points):** This detailed report, a 2-page summary for the presentation, and the source code hosted on GitHub.

3. Key Features

The application distinguishes itself through the following implementation details:

- **Three Difficulty Levels:** Users can dynamically toggle between Easy (1-50), Medium (1-100), and Hard (1-500). The target number resets instantly upon difficulty change.
- **Persistent User System:** Credentials are not hard-coded; they are stored and retrieved from a local file, ensuring users can return to their profiles.
- **Live Leaderboard:** A dedicated "Score History" window displays recorded attempts, allowing players to track their performance over time.

- **Modern UI/UX:** Built with a dark-themed aesthetic, featuring responsive buttons, hover effects, and keyboard support (Enter key for guessing).

4. System Architecture

The project follows the Separation of Concerns principle, organized into four primary classes to maintain "Clean Code" standards:

4.1. Launcher.java

Acts as the entry point to bypass JavaFX runtime issues, ensuring the application starts correctly across different environments.

4.2. Main.java

The "Director" of the application. It handles:

- **Scene Switching:** Seamless transitions between the Login/Register and Game scenes.
- **UI Components:** Layout management using VBox, HBox, and styled Nodes.
- **Event Handling:** All button clicks and user inputs are processed here.

4.3. UserManager.java

Handles the File Processing and Authentication logic:

- **I/O Operations:** Uses BufferedReader and BufferedWriter to interact with users.txt and scores.txt.
- **Data Structure:** Utilizes a HashMap for fast credential verification during login.

4.4. GameLogic.java

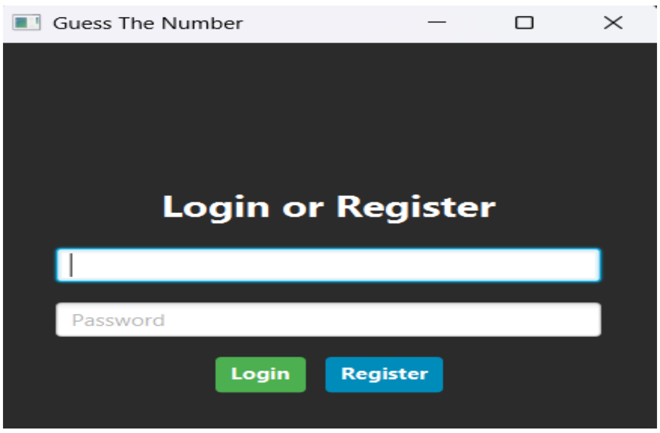
The "Engine" of the game. It is completely independent of the UI, handling:

- **Random number generation** based on the selected range.
- **Comparison logic** to determine if a guess is "Too High," "Too Low," or "Correct."

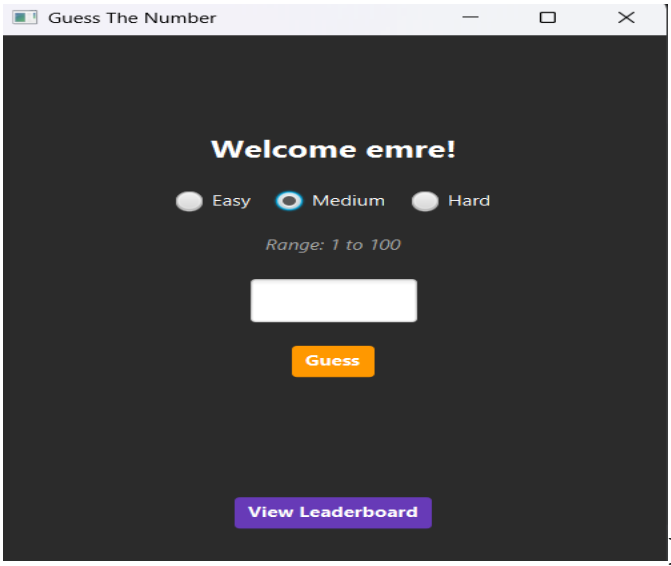
5. Visual Evidence

The following components demonstrate the successful execution of the application:

Login & Registration: Secure entry point validating user existence and password accuracy.



Main Game Interface: Interactive HUD showing current user, range information, and real-time feedback.



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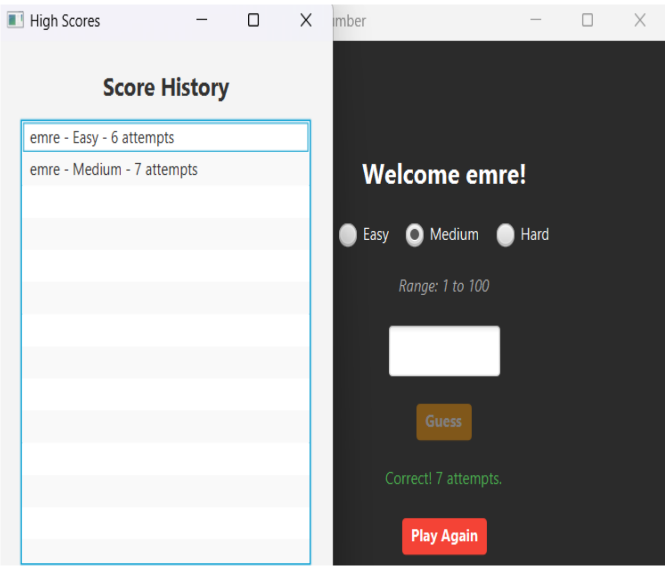
- **Leaderboard Window:** Proof of file I/O operations where game results are successfully retrieved from the local drive.



6. Installation and Setup

To deploy and review this project:

1. Clone the repository from the GitHub link provided above.
2. Import the project into an IDE (IntelliJ IDEA recommended) as a Maven project to load dependencies from pom.xml.
3. Ensure the JavaFX SDK is correctly configured in the project structure.
4. Run Launcher.java to start the application.

7. Conclusion

This project fulfills the requirements for a visual JavaFX application with integrated file processing and authentication systems. Beyond meeting the technical criteria, the "Guess The Number" game provides a scalable foundation for further gamification features.